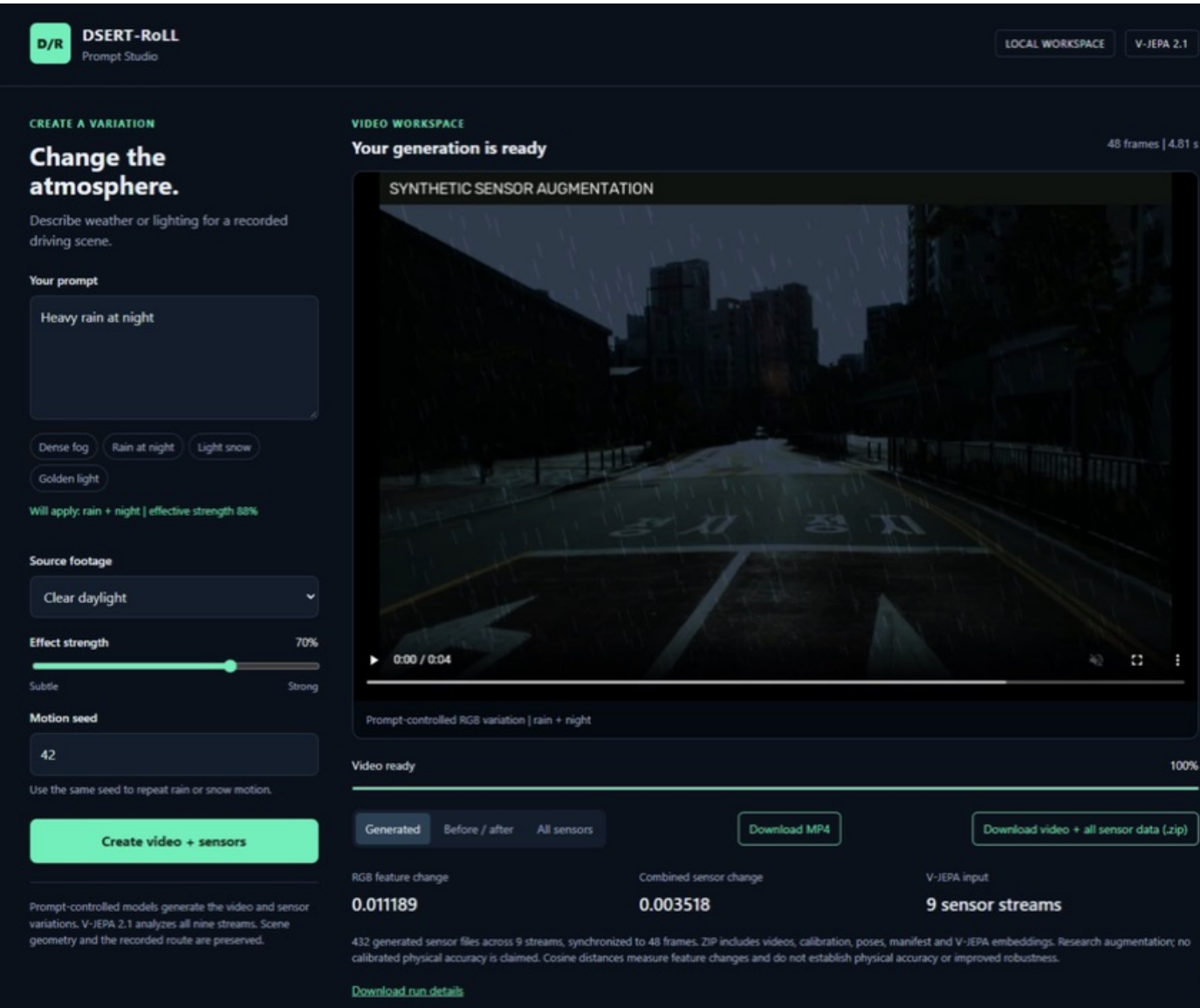


# One prompt. Video + nine streams.

A local interface for DSERT-RoLL weather and lighting augmentation, with V-JEPA 2.1 representation analysis.



Live remote UI: the verified **Heavy rain at night** run, with its native sensor ZIP download and V-JEPA measurements.

|                   |                                    |                     |                         |
|-------------------|------------------------------------|---------------------|-------------------------|
| 48 frames         | 9 streams                          | 432 sensor files    | 150.24 seconds          |
| 4.806-second clip | RGB, events, thermal, LiDAR, radar | 48 files per stream | Measured remote runtime |

**How it works:** project code creates weather and sensor variations. V-JEPA 2.1 encodes the source and generated visualizations. It does not decode prompts into pixels or native sensor arrays.

# Create and compare a variation

Open <http://127.0.0.1:8765> on the remote computer. The server is local to that computer.

- 1. Choose the source.** Select Clear daylight.
- 2. Enter the prompt.** Use Heavy rain at night. Other supported effects: fog, rain, snow, night and warm light.
- 3. Set the controls.** Strength 0.7; seed 42. The word heavy raises effective strength to 0.875.
- 4. Generate.** Select Create video + sensors. Allow the job to finish; the server runs one job at a time.
- 5. Inspect and download.** Use Generated, Before / after and All sensors. Download the full ZIP for native data.

**DSERT-RoLL**  
Prompt Studio

LOCAL WORKSPACE V-JEPA 2.1

**CREATE A VARIATION**

**Change the atmosphere.**  
Describe weather or lighting for a recorded driving scene.

**Your prompt**  
Heavy rain at night

Dense fog Rain at night Light snow  
Golden light

Will apply: rain + night | effective strength 88%

**Source footage**  
Clear daylight

**Effect strength** 70%  
Subtle Strong

**Motion seed**  
42

Use the same seed to repeat rain or snow motion.

**Create video + sensors**

Prompt-controlled models generate the video and sensor variations. V-JEPA 2.1 analyzes all nine streams. Scene geometry and the recorded route are preserved.

**VIDEO WORKSPACE**  
Your generation is ready 48 frames | 4.81 s

RECORDED SOURCE SYNTHETIC SENSOR AUGMENTATION

0:00 / 0:04

Recorded source on the left | synthetic variation on the right

Video ready 100%

Generated Before / after All sensors Download MP4 Download video + all sensor data (zip)

RGB feature change 0.011189 Combined sensor change 0.003518 V-JEPA input 9 sensor streams

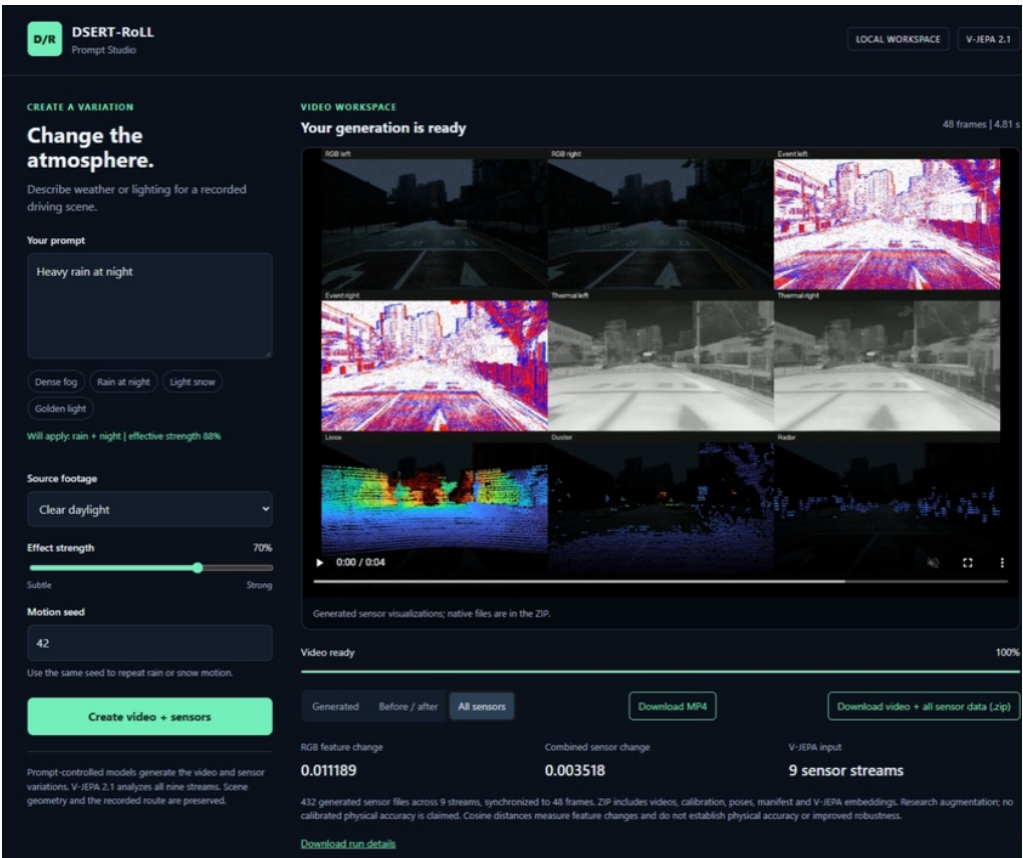
432 generated sensor files across 9 streams, synchronized to 48 frames. ZIP includes videos, calibration, poses, manifest and V-JEPA embeddings. Research augmentation; no calibrated physical accuracy is claimed. Cosine distances measure feature changes and do not establish physical accuracy or improved robustness.

[Download run details](#)

Before / after: recorded daylight on the left, synthetic rain and darkness on the right. Geometry and the recorded route remain from the dataset.

# Inspect all generated sensor streams

The All sensors view previews the generated files. The ZIP contains the underlying arrays, images and provenance.



| Family / streams       | Native format       | What changes                                   |
|------------------------|---------------------|------------------------------------------------|
| RGB / left + right     | 960 x 600 PNG       | Lighting, haze, precipitation                  |
| Events / left + right  | uint32 NPZ, N x 4   | Event thinning and injected noise              |
| Thermal / left + right | Non-radiometric PNG | Intensity and contrast                         |
| LiDAR / Livox + Ouster | Structured NPY      | Range noise, dropout, attenuation, backscatter |
| Radar / one stream     | Structured NPY      | Range, power and Doppler noise                 |

**Units and calibration:** event timestamps retain source-local units. Thermal values are display intensities, not temperatures. Radar fields retain source conventions. Original frame timestamps and poses are preserved.

The verified bundle contains 88,286,126 events and 3,828,874 points. All 48 frames in all nine streams changed.

# Set up a reproducible checkout

Tested: Ubuntu 24.04 under WSL, Python 3.11, PyTorch 2.5.1 + CUDA 12.1, one Quadro RTX 5000 with 16 GB VRAM.

**Prerequisites.** Install Git, FFmpeg and Python 3.11 with venv support. Clone the private repository using your own authorized GitHub access. Setup downloads several GB of public model and dataset assets.

Install FFmpeg on Ubuntu

```
sudo apt-get install ffmpeg
```

Prepare environment, pinned model source and data

```
bash scripts/setup.sh
```

Start the local studio

```
CUDA_VISIBLE_DEVICES=0 bash prompt-studio/start.sh
```

Check sensor transforms after data preparation

```
.venv311/bin/python prompt-studio/test_sensor_models.py
```

Check outputs for a completed run

```
.venv311/bin/python prompt-studio/verify_sensor_app.py JOB_ID
```

Use GPU index **1** on the previously tested remote workstation. The current model code requires CUDA; it does not combine multiple GPUs. The first run may take longer while loading the checkpoint.

**Output root:** prompt-studio/jobs/<job-id>/

| File / directory               | Purpose                                            |
|--------------------------------|----------------------------------------------------|
| video.mp4 / comparison.mp4     | Generated camera video and source comparison       |
| synthetic_sensors.mp4          | Preview of all nine generated streams              |
| sensor_data.zip / sensor_data/ | Native files, metadata, calibration and provenance |
| embeddings.npz / metrics.json  | V-JEPA features, parameters, distances and timing  |
| validation.json                | Schema, timestamp and integrity checks             |

**Troubleshooting:** inspect prompt-studio/server.log for startup errors. Confirm FFmpeg and the selected CUDA GPU are available. Re-run setup if the model or data is missing. Keep this research server bound to localhost.

# What was verified

The example is a measured remote experiment. Its outputs are research augmentations rather than a calibrated simulator.

| Validation            | Recorded result                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------|
| Source data integrity | Original source hashes unchanged                                                            |
| Native files          | 432 files present with expected schemas and dimensions                                      |
| Temporal organization | 48 strictly ordered frame timestamps                                                        |
| Video integrity       | Complete FFmpeg decoding passed                                                             |
| Archive integrity     | ZIP CRC validation passed; approximately 586 MB                                             |
| V-JEPA coverage       | All nine visualized streams encoded; 16 sampled frames per stream, 768-dimensional features |

**Interpretation.** RGB cosine distance was 0.011189; the combined sensor feature distance was 0.003518. These quantify changes in visual representation. They do not establish physical realism, sensor accuracy, or downstream robustness.

**Research boundary.** Weather models are sensor-specific heuristics. They do not share a calibrated particle or ray-tracing world. The system preserves the source scene, route, poses and timestamps; it does not create new GPS/IMU trajectories or arbitrary objects.

**Pinned model source**

facebookresearch/vjepa2  
204698b45b3712590f06245fbfba32d3be539812  
Backbone: vjepa2\_1\_vit\_base\_384 (V-JEPA 2.1). The included patch points the loader at Meta's official checkpoint host.

**Primary sources and project**

[Project repository](#)  
[Meta V-JEPA source](#)  
[DSERT-RoLL dataset](#)  
[Official dataset code](#)  
Dataset revisions are pinned in the selection JSON files. Model and dataset licenses remain those of their upstream owners. Weights, full recordings, credentials and generated job bundles are excluded from the code repository.